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Experiment No.	9

Aim:	To develop and evaluate a facial image classification model using UTKFace, perform data preprocessing and exploratory data analysis, evaluate performance across demographic subgroups, and investigate prediction errors using explainability techniques.
Experiment 9	
Theory:	This experiment applies transfer learning to classify UTKFace images into five age groups. An ImageNet-pretrained ResNet-18 with a frozen backbone and a new linear head was trained. Since overall accuracy hides class imbalance, balanced accuracy, macro F1, subgroup audits and LIME/Grad-CAM explanations were also used.
Results:	After removing 3 malformed and 390 duplicate files, 23,315 images remained. The model achieved 71.73% test accuracy but only 56.17% balanced accuracy and 0.5965 macro F1. Young adult recall was 0.92, while Teenager recall was 0.19; most errors pushed Teenagers and Middle-aged adults into Young adult.
Conclusion:	Transfer learning gave a quick, reasonable baseline, but the 10:1 class imbalance made the model default to Young adult. Subgroup accuracy differences largely reflect differing age compositions, not proven bias. LIME and Grad-CAM show associations, not causation, and these results imply no clinical or genetic-disorder diagnostic validity.